Coffe Quality Analysis

Problem Description :

Background: Coffee is one of the most popular beverages consumed worldwide, and its quality greatly influences the overall coffee-drinking experience. The quality of coffee is determined by various factors such as flavor, aroma, acidity, body, balance, and overall taste. Coffee producers and roasters are constantly striving to improve the quality of their products to meet consumer preferences and market demands.

Objective: The main objective of the "Coffee Quality Analysis" project is to analyze and predict the quality of coffee based on various attributes. By leveraging data on coffee beans from different countries and their associated sensory ratings, we aim to gain insights into what factors contribute to high-quality coffee and build a predictive model that estimates the overall quality of coffee based on its characteristics.

Dataset Information: The dataset used in this project contains information about coffee beans from various countries and their corresponding sensory ratings, including aroma, flavor, acidity, body, balance, and overall cup points. Additional attributes such as the processing method, altitude, coffee age, and color are also provided. The dataset is a valuable resource for coffee producers, roasters, and enthusiasts to understand the factors that influence coffee quality and make informed decisions in the coffee production process.

Dataset Columns:

1. ID: Unique identifier for each coffee entry
2. Country of Origin: The country where the coffee beans were grown
3. Processing Method: The method used to process the coffee beans (e.g., washed, pulped natural, natural)

4. Altitude: The altitude at which the coffee beans were grown
5. Aroma: Sensory rating for the aroma of the coffee
6. Flavor: Sensory rating for the flavor of the coffee
7. Aftertaste: Sensory rating for the aftertaste of the coffee
8. Acidity: Sensory rating for the acidity of the coffee
9. Body: Sensory rating for the body of the coffee
10. Balance: Sensory rating for the balance of the coffee
11. Overall: Overall sensory rating for the coffee
12. Total Cup Points: Aggregate score representing the overall quality of the coffee
13. Moisture Percentage: Percentage of moisture in the coffee beans
14. Category One Defects: Number of category one defects in the coffee beans
15. Quakers: Number of quaker beans in the coffee
16. Category Two Defects: Number of category two defects in the coffee beans
17. Coffee Age: Age of the coffee from harvest to expiration
18. Color: Color of the coffee beans
19. Harvest Year: Year when the coffee beans were harvested
20. Expiration: Expiration date of the coffee

Approach:

- Data Exploration and Preprocessing: We will start by exploring the dataset, checking for any missing values or duplicate rows, and performing data cleaning and transformation as needed.
- Data Visualization: Visualizations like histograms, choropleth maps, and bar plots will be used to gain insights into the distribution of attributes and the relationships between different variables.
- Feature Engineering: We will extract relevant information from the columns, such as calculating 'Coffee Age' based on the 'Harvest Year' and 'Expiration' columns.
- Correlation Analysis: To understand the relationship between the attributes and the target variable 'Total Cup Points,' we will perform correlation analysis.
- Predictive Modeling: Finally, we will build a predictive model using regression techniques to estimate the overall quality of coffee based on the given attributes.

Expected Outcome: The project's expected outcome is a predictive model that can estimate the overall quality of coffee based on its sensory attributes and other features. This model can be used by coffee producers and roasters to assess the quality of their

coffee beans, identify areas for improvement, and make data-driven decisions to enhance the overall coffee-drinking experience for consumers.

Possible Framework :

Step 1: Data Loading and Exploration

- Import the necessary libraries for data analysis, visualization, and modeling.
- Load the coffee quality dataset into a pandas DataFrame.
- Check the first few rows of the dataset to get an overview of the data.
- Check for any duplicate rows in the dataset and handle them if present.
- Check for missing values in the dataset and handle them appropriately.
- Explore the unique values in each column to understand the data better.

Step 2: Data Preprocessing

- Perform data cleaning and transformation on various attributes:
- Clean and transform the 'Processing Method' column to standardize the processing categories.
- Handle missing altitude values and clean the 'Altitude' column by averaging altitude ranges.
- Extract relevant information from the 'Harvest Year' column to calculate the 'Coffee Age.'
- Drop irrelevant columns that do not contribute to the analysis or modeling process.

Step 3: Data Visualization

- Generate histograms to visualize the distribution of numerical attributes such as aroma, flavor, acidity, and more.
- Use choropleth maps to visualize average total cup points, coffee shelter life, and altitude based on the country of origin.
- Create bar plots to visualize average total cup points and altitude by country.
- Visualize the correlation between attributes and the target variable using heatmaps.

Step 4: Feature Engineering and Correlation Analysis

- Prepare the data for modeling by encoding categorical variables like 'Processing Method,' 'Country of Origin,' and 'Color.'

- Scale numerical attributes using Min-Max scaling to bring them to a similar scale for modeling.
- Perform correlation analysis to understand the relationship between the attributes and the target variable 'Total Cup Points.'

Step 5: Building the Predictive Model

- Split the dataset into training and testing sets to evaluate the model's performance.
- Build a pipeline for preprocessing the data, including one-hot encoding for categorical features and scaling for numerical features.
- Choose a suitable regression model for the prediction task, such as Random Forest Regressor or Gradient Boosting Regressor.
- Train the chosen model on the training data and evaluate its performance using Mean Squared Error (MSE) on the test data.
- Visualize the residuals and prediction error to assess the model's performance.

Step 6: t-SNE Visualization

- Apply t-SNE (t-distributed Stochastic Neighbor Embedding) to reduce the high-dimensional data to 2D.
- Visualize the 2D representation of data points colored by their 'Total Cup Points' using a scatter plot.

Step 7: Conclusion and Results

- Summarize the findings from the data exploration, visualization, and modeling.
- Discuss the insights gained from the project, such as identifying factors that contribute to high-quality coffee.
- Present the predictive model's performance and discuss its accuracy in estimating coffee quality.
- Conclude the project with suggestions for potential improvements or future directions for coffee quality analysis.

Step 8: Additional Insights (Optional)

- Explore more advanced visualizations, such as interactive plots using Plotly, to gain deeper insights into the data.

- Apply dimensionality reduction techniques like PCA (Principal Component Analysis) and visualize the results.
- Experiment with different regression models and compare their performance to choose the best one for the task.

Step 9: Documentation and Presentation

- Organize the code and documentation for easy readability and understanding.
- Create a presentation or report summarizing the project's process, findings, and results.
- Include visualizations and graphs to support the analysis and convey the key insights effectively.

Step 10: Conclusion

- Review the entire project and its objectives.
- Reflect on the achieved results and the impact of the predictive model on coffee quality assessment.
- Discuss potential real-world applications and the significance of the project in the coffee industry.
- Suggest possible future work and improvements for further enhancing the coffee quality analysis model.

Code Explanation :

*If this section is empty, the explanation is provided in the .ipynb file itself.

Step 1: Data Loading and Exploration

- The code starts by importing the required libraries for data analysis, visualization, and modeling. These libraries include pandas, numpy, matplotlib, seaborn, and plotly.
- Next, the code loads the coffee quality dataset from a CSV file into a pandas DataFrame using the pd.read_csv() function.
- The first few rows of the dataset are displayed using the df.head() function to get an overview of the data.
- The code then checks for any duplicate rows in the dataset using the df.duplicated() function and handles them if present.
- It also checks for missing values in the dataset using the df.isnull().sum() function and calculates the percentage of missing data for each column.
- The code then performs some data preprocessing tasks like mapping and handling missing values in the 'Processing Method' and 'Altitude' columns.

Step 2: Data Visualization

- In this step, the code generates histograms to visualize the distribution of various numeric attributes in the dataset such as aroma, flavor, acidity, etc. The sns.histplot() function from the seaborn library is used to create the histograms.
- The code also uses choropleth maps from Plotly to visualize the average total cup points, coffee shelter life, and altitude based on the country of origin. These maps help in understanding how these attributes vary across different countries.
- Bar plots are created to visualize the average total cup points and altitude for each country, which provides insights into the overall coffee quality across different regions.

Step 3: Data Preprocessing

- In this step, the code performs data cleaning and transformation on various attributes in the dataset.

- The 'Processing Method' column is standardized by mapping similar processing methods into common categories. Missing values in this column are filled with a default value.
- The 'Altitude' column is cleaned by averaging altitude ranges and filling missing values with appropriate estimates.
- The 'Harvest Year' column is used to calculate the 'Coffee Age' by subtracting the harvest year from the expiration date. Irrelevant columns that do not contribute to the analysis or modeling process are dropped.

Step 4: Feature Engineering and Correlation Analysis

- The code prepares the data for modeling by encoding categorical variables like 'Processing Method,' 'Country of Origin,' and 'Color' using label encoding.
- Numerical attributes are scaled using Min-Max scaling to bring them to a similar scale for better model performance.
- Correlation analysis is performed to understand the relationship between attributes and the target variable, 'Total Cup Points.' Heatmaps are used to visualize the correlation matrix, making it easier to interpret the relationships.

Step 5: Building the Predictive Model

- The dataset is split into training and testing sets to evaluate the model's performance.
- A pipeline is constructed to preprocess the data, including one-hot encoding for categorical features and scaling for numerical features.
- The code chooses a suitable regression model (Random Forest Regressor) for the prediction task.
- The chosen model is trained on the training data, and its performance is evaluated using Mean Squared Error (MSE) on the test data.

Step 6: t-SNE Visualization

- In this step, the code applies t-SNE (t-distributed Stochastic Neighbor Embedding) to reduce the high-dimensional data to 2D.
- t-SNE is a dimensionality reduction technique used for visualizing high-dimensional data in a lower-dimensional space.
- The 2D representation of data points is then visualized using a scatter plot, with the points colored by their 'Total Cup Points.'

Step 7: Conclusion and Results

- The code concludes the analysis by summarizing the findings from the data exploration, visualization, and modeling.
- Insights gained from the project are discussed, such as identifying factors that contribute to high-quality coffee.
- The predictive model's performance is presented, and its accuracy in estimating coffee quality is discussed.
- The project is concluded with suggestions for potential improvements or future directions for coffee quality analysis.
- The code demonstrates a comprehensive workflow for analyzing coffee quality data, starting from data loading and exploration to data visualization, preprocessing, modeling, and concluding with valuable insights and suggestions.

Future Work :

Analyzing coffee quality is an exciting area with vast potential for further exploration and improvement. Here are some future work suggestions to enhance the project:

Step 1: Data Enhancement

- Acquire Additional Data: Obtain a larger and more diverse dataset that includes a broader range of coffee types, origins, and processing methods. More data can lead to better model performance and more reliable insights.

Step 2: Feature Engineering

- Extract New Features: Explore the dataset to identify additional relevant features that might influence coffee quality. For example, weather conditions during coffee cultivation or specific coffee bean varieties could be potential features.

Step 3: Advanced Visualization

- Interactive Dashboards: Create interactive dashboards using tools like Plotly Dash or Tableau to allow users to explore coffee quality trends and attributes easily. Interactive visualizations can provide a more engaging user experience.

Step 4: Advanced Modeling

- Ensemble Methods: Implement ensemble models like stacking or boosting, combining multiple models to improve prediction accuracy and robustness.

Step 5: Hyperparameter Tuning

- Grid Search: Perform an exhaustive grid search for hyperparameter tuning of the regression model to optimize its performance.

Step 6: Model Evaluation Metrics

- Evaluate Different Metrics: Besides Mean Squared Error (MSE), consider other evaluation metrics such as R-squared (R2), Mean Absolute Error (MAE), or root-mean-square error (RMSE) to better understand the model's performance.

Step 7: Explainable AI

- Model Interpretability: Implement explainable AI techniques like SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-agnostic Explanations) to interpret the model's predictions and understand which features contribute most to the coffee quality.

Step 8: Deployment

- Web Application: Develop a user-friendly web application where users can input coffee attributes, and the model predicts the quality and provides insightful visualizations.

Step-by-Step Guide to Implement Future Work

Step 1: Data Enhancement

- Search for publicly available coffee quality datasets or collaborate with coffee industry stakeholders to obtain more comprehensive and diverse data.
- Ensure the new data is well-structured and contains relevant attributes for coffee quality analysis.

Step 2: Feature Engineering

- Carefully explore the new dataset to identify potential additional features that might impact coffee quality.
- Transform and preprocess the new features to match the existing data format.

Step 3: Advanced Visualization

- Utilize tools like Plotly Dash or Tableau to create interactive dashboards for coffee quality visualization.
- Implement user-friendly controls and filters to allow users to explore specific aspects of coffee quality easily.

Step 4: Advanced Modeling

- Research and experiment with different ensemble methods like stacking or boosting to combine multiple models.
- Assess the performance of ensemble models and compare them with the existing random forest regression model.

Step 5: Hyperparameter Tuning

- Use GridSearchCV or RandomizedSearchCV to explore the hyperparameter space of the regression model.
- Identify the best hyperparameters that yield optimal model performance.

Step 6: Model Evaluation Metrics

- Evaluate the regression model using various metrics such as R-squared, MAE, and RMSE.
- Compare the results with the existing MSE metric to gain a more comprehensive understanding of the model's performance.

Step 7: Explainable AI

- Implement SHAP or LIME to interpret the predictions of the regression model.
- Visualize the feature importance and how each attribute contributes to the coffee quality prediction.

Step 8: Deployment

- Develop a web application using frameworks like Flask or Django.
- Integrate the trained model into the application to make real-time predictions based on user input.
- Design an intuitive user interface to display predictions and visualizations to end-users.

By following this step-by-step guide, you can enhance the Coffee Quality Analysis project with improved data, advanced modeling, and interactive visualizations, making it a valuable tool for coffee industry professionals and enthusiasts alike.

Concept Explanation :

Alright, imagine you're hosting a grand coffee-bean sorting party, and you've got tons of coffee beans from different countries. Your goal is to separate the best quality coffee beans from the not-so-great ones. But, there's just too much work for you to do alone, and you need some clever friends to help you out.

In comes the "Random Forest" crew! ⍰⍰⍰

Each member of the Random Forest crew is a "Decision Tree," and they are all wearing different outfits (features) to identify the best coffee beans. Some wear hats, some wear glasses, and others wear fancy shoes – representing the different attributes of coffee beans like aroma, flavor, acidity, and more.

Let's say you have a sample of coffee beans with known quality ratings. Each bean is unique and has its own combination of attributes, just like people in real life. Your crew of decision trees will each look at a coffee bean and cast their votes on whether it's high-quality or not.

Now, each decision tree is a bit quirky and may not be perfect. Some of them may have a bias towards certain attributes. For example, one tree might really love the flavor, while another focuses on the bean's altitude. But that's fine, we love quirky friends, right? ⍰

After all the decision trees have voted, you gather their decisions and let them throw a party (that's the "Random" part!). Each decision tree gets to party and celebrate their own decision. ⍰

But, you're a wise host and want to make the best decision for the party (i.e., find the best coffee beans!). So, you take a majority vote – whichever decision gets the most votes, wins! This way, you balance out the quirks and biases of individual decision trees. They all celebrate together with their favorite attributes.

And voilà! ⍰ You have successfully sorted the best coffee beans from the rest, thanks to your trusty Random Forest crew!

Now, you might wonder, why did we call it a "forest"? Well, because the more decision trees you invite to your party, the better the results! Each tree has a unique perspective, and the more you have, the more diverse and accurate the final decision becomes.

So, the Random Forest is like hosting a fun, diverse party where each quirky friend brings their insights to the table. In the end, you get a fantastic coffee-bean sorting solution that is robust, reliable, and full of flavor – just like a perfect cup of coffee! ☕

Now go ahead and impress your friends at the next coffee tasting event with your magical Random Forest knowledge! Have fun sorting those coffee beans! ☕

Exercise Questions :

1. **What is the purpose of the Coffe Quality Analysis project, and what kind of data does it use?**
   - The purpose of the Coffee Quality Analysis project is to analyze and understand the factors that contribute to the quality of coffee beans. The project uses a dataset that includes various attributes of coffee beans, such as aroma, flavor, acidity, altitude, processing method, and more.
2. **Why is it essential to preprocess the data before building the machine learning model?**
   - Data preprocessing is crucial because it helps in cleaning, transforming, and organizing the data to make it suitable for the machine learning model. Preprocessing includes handling missing values, scaling numerical features, and encoding categorical variables so that the model can effectively learn from the data.
3. **Explain the role of the Random Forest algorithm in this project.**
   - The Random Forest algorithm is used in this project to build a predictive model that can determine the quality of coffee beans based on their attributes. It works by creating multiple decision trees and combining their outputs to make a final decision. The Random Forest provides a robust and accurate solution, considering the diversity of opinions from individual decision trees.
4. **How does the project visualize the relationships between attributes and coffee quality?**
   - The project uses various visualization techniques like histograms, choropleth maps, bar plots, and scatter plots to show the relationships between different attributes and the overall coffee quality. These visualizations help understand patterns and correlations between variables.
5. **Can you explain the concept of t-SNE in simple terms and how it's used in the project?**
   - t-SNE (t-distributed Stochastic Neighbor Embedding) is a dimensionality reduction technique that helps visualize high-dimensional data in a lower-dimensional space. In the project, t-SNE is used to reduce the feature space of coffee bean attributes to two dimensions so that the data can be plotted

on a 2D scatter plot. This allows us to visually observe clusters and patterns in the data.

6. **What are the steps involved in building the machine learning model for predicting coffee quality?**

The steps involved in building the machine learning model are:

- Data preprocessing: Cleaning, transforming, and scaling the data.
- Feature selection: Identifying the relevant attributes for the model.
- Model selection: Choosing the appropriate algorithm (Random Forest Regressor) for the regression task.
- Training: Using the training data to teach the model to make predictions.
- Evaluation: Assessing the model's performance on the test data using metrics like Mean Squared Error.

7. **How does the project handle missing values and outliers in the dataset?**
   - The project handles missing values by either filling them with appropriate values (e.g., mean, median) or replacing them with specific default values where applicable. For outliers, the project may use techniques like capping or transformations to reduce their impact on the model.
8. **Can you explain the significance of the correlation matrix and how it helps in the analysis?**
   - The correlation matrix shows the relationships between different attributes in the dataset. It helps identify which attributes have a strong positive or negative correlation with coffee quality. This information is crucial in understanding which features are most influential in determining the coffee quality.
9. **How does the project handle categorical features, and why is one-hot encoding used?**
   - Categorical features like "Processing Method," "Country of Origin," and "Color" are encoded using one-hot encoding. This is necessary because machine learning algorithms typically work with numerical data. One-hot encoding transforms categorical variables into binary vectors, allowing the model to treat each category independently and avoid introducing ordinal relationships.

10. How do you evaluate the performance of the machine learning model?

- The performance of the machine learning model is evaluated using Mean Squared Error (MSE), which measures the average squared difference between the predicted and actual values of coffee quality. Lower MSE values indicate better model performance, as they signify less error in the predictions. Residual plots are also used to visually inspect the model's prediction errors.